

Reported pediatric cancers in Ladera Ranch: what the public evidence does and does not show

Independent public-source research · Generated 2026-07-18 · Version 0.5.0 · Hypothesis-neutral · Aggregate-only

This report does not provide medical advice and does not establish that any pesticide, property, organization, employer, school, water provider, government agency, or other party caused any illness. Publicly reported health events may not have been independently medically verified. Geographic and temporal overlap does not establish exposure or causation. Formal conclusions require authorized epidemiological analysis, verified medical information, exposure assessment, toxicological review, and independent scientific evaluation. Health data here is aggregate-only; no individual is identified.



1. Executive summary

No agency has declared a cluster or identified a cause. An official multi-agency review — Orange County Health Care Agency, the California Cancer Registry, UC Irvine, and the OC Agricultural Commissioner — is underway; its initial pass "did not find a particular pattern," and findings remain pending. A federal EPA investigation has been *requested* by the First Assistant U.S. Attorney but not confirmed opened, and no EPA response was found.

Every pesticide-specific causal claim in circulation is advocacy- or attorney-sourced (grade C) — including the "17 pesticides in June" figure and the naming of glufosinate. No regulator classifies glufosinate as a carcinogen; its EU non-renewal was on reproductive-toxicity grounds, not cancer.

The single most important scientific fact is hypothesis-neutral: Ewing sarcoma is strongly ancestry-patterned through germline biology — roughly nine times more common in people of European ancestry than African ancestry. A community that is 63.6% non-Hispanic white therefore carries an *elevated baseline expectation*, which must be modelled before any excess is attributed to the environment.

The strongest testable environmental lead is not active spraying. It is *legacy agricultural soil residue*: the community was built on former citrus, grain and nursery land, and neighbouring former-farm school sites in the state's own database repeatedly test positive for DDT, toxaphene, chlordane and arsenic — yet the footprint itself appears never to have been tested.

Bottom line. Under transparent assumptions the *reported* count exceeds statistical expectation, which is precisely why the pattern **warrants investigation**. It is not proof of a cluster: the counts are media/attorney-reported rather than registry-verified, the boundary was drawn around the observed cases, and the numbers are small enough that one or two cases move the result dramatically. **The available evidence does not yet establish causation.**

2. Study area and environmental context

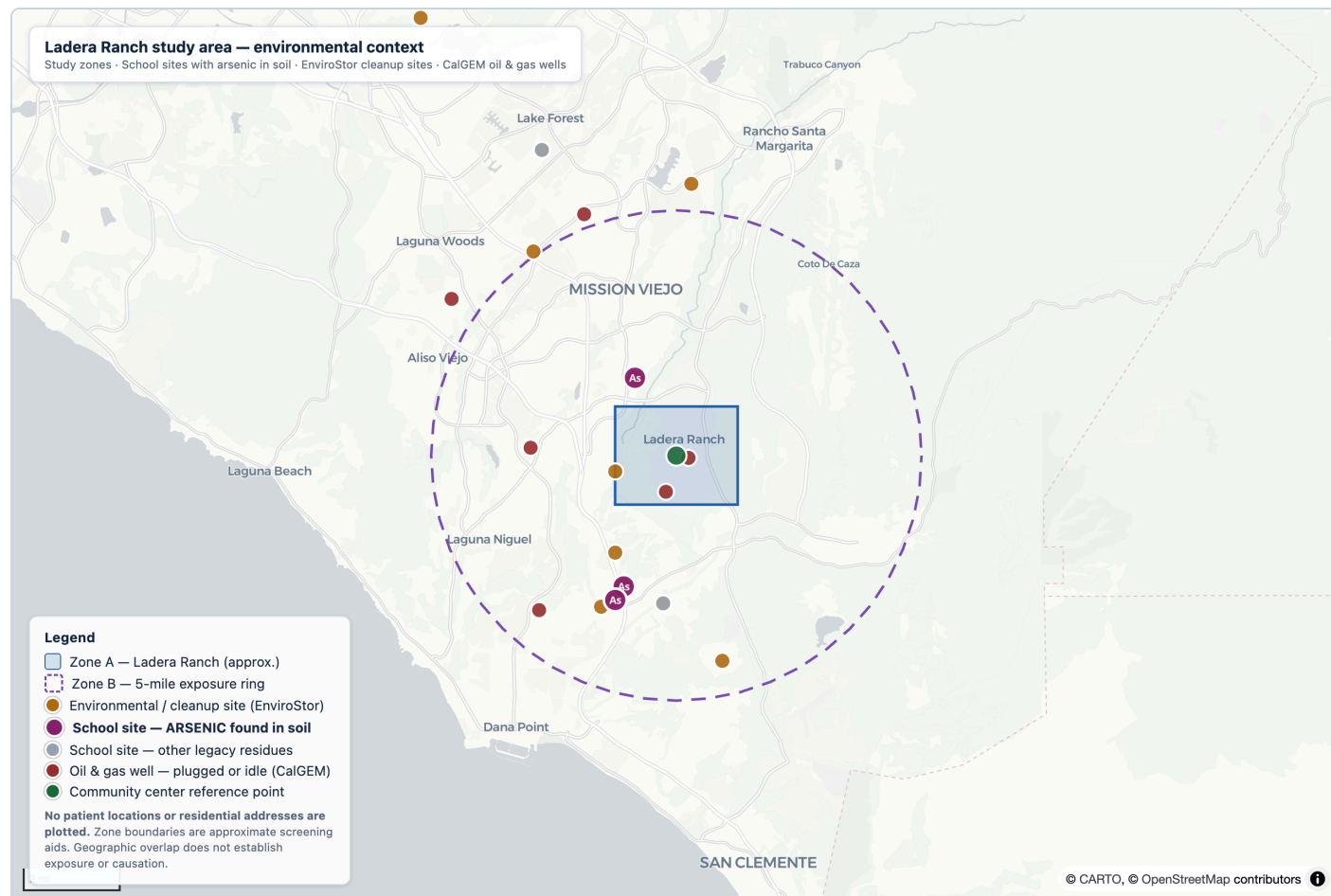


Figure 1. Study zones with DTSC EnviroStor cleanup sites and CalGEM oil & gas wells. Zone A (solid) is an approximate ~2.5 × 2 mile screening boundary centred on the Ladera Ranch CDP centroid; Zone B (dashed) is a 5-mile exposure ring. Site and well points use real database coordinates. **No patient locations or residential addresses are plotted.** Basemap © CARTO, © OpenStreetMap contributors.

3. What relationships the data can — and cannot — establish

On "correlation". This platform does not compute a correlation between pesticide use and cancer, and no such figure should be quoted from it. Doing so would require case-level data the platform does not hold, and an area-level correlation would be an *ecological fallacy* — an association between aggregates that does not

imply anything about individuals. What the public data *can* support is described below: spatial co-location, distance, timing overlap, and an observed-versus-expected comparison under stated assumptions. Each is a screening signal, not evidence of cause.

What can be shown

- Distance from the community to documented environmental sites and wells.
- Whether land-use eras overlap the residency and diagnosis window.
- Observed vs expected case counts under explicit, varied assumptions.
- Which chemicals are actually reported in regional landscape use.
- What the peer-reviewed literature does and does not support.

What cannot be shown

- That any individual was exposed to anything.
- Any dose, or contact between a person and a chemical.
- That a statistically valid cluster exists.
- Any causal link between an exposure and an illness.
- A meaningful pesticide↔cancer correlation coefficient.

4. Spatial relationships

4.1 Environmental and cleanup sites, by distance

SITE	TYPE / DATABASE	CONTAMINANTS	MILES
Blue Diamond Materials Plant	Evaluation (asphalt) EnviroStor	(unspecified)	1.3
Carl Hankey Elementary School	School Investigation (former orchard + row crops) EnviroStor	Arsenic, Lead, Methane	1.8
Meggitt (OC) Inc / former Endevco	Tiered Permit (electronics mfg) EnviroStor	Lead	2.3
Plant Depot School Site	School Investigation (ag services) EnviroStor	Arsenic, Nitrate	2.9
Ambuehl Elementary School	School Investigation EnviroStor	DDD, DDE, DDT, Toxaphene	3.0
San Juan Elementary School	School Investigation (former orchard) EnviroStor	Arsenic, Chlordane, DDD, DDE, DDT, Lead	3.2
SOLAG Disposal Site	Voluntary Cleanup (junkyard/fuel) EnviroStor	Arsenic, Cr, Lead, PAHs, Beryllium	3.5
Prima Deshecha Landfill	Active Class III landfill (SWIS 30-AB-0019) CalRecycle/SD Water Board	Landfill gas (methane), leachate; groundwater & LFG monitored	4.3
Alicia Towne Plaza	Voluntary Cleanup (dry cleaner) EnviroStor	PCE	5.1
Polo Cleaners	Voluntary Cleanup EnviroStor	1,1,1,2-tetrachloroethane; 1,1,2-TCA	5.6
El Toro High School	School Investigation (former row crops) EnviroStor	PCBs, TPH-diesel	6.8

SITE	TYPE / DATABASE	CONTAMINANTS	MILES
El Toro MCAS (former)	Federal Superfund / NPL EPA/DTSC	TCE (VOC groundwater plume migrated >3 mi off-base)	10.3
Capistrano Test Site (Northrop Grumman/former TRW)	Former rocket/laser test range (~2700 ac) EnviroStor/RWQCB	Potential perchlorate/NDMA (by analogy; site-specific results NOT verified)	10.5

Pattern worth noting. The nearest former-agricultural school sites — Plant Depot (arsenic, nitrate), Ambuehl (DDT, toxaphene) and San Juan Elementary (arsenic, chlordane, DDT) — lie roughly 2.9–3.2 miles away and carry exactly the legacy residue profile expected on former California citrus and row-crop land. Ladera Ranch was built on comparable land. This is a *testable hypothesis*, not a finding.

4.2 Oil & gas wells (CalGEM)

WELL (CALGEM)	STATUS	MILES FROM CENTROID
Citizens National Trust B-1 / Exxon Mobil	Plugged, Dry Hole	0.25
O'Neill #1 / Union Oil	Plugged, Dry Hole	0.77
Shumaker #1 / Conoco	Plugged, Dry Hole	2.98
Norswing & Halvorson #1	Idle, Oil & Gas	4.22
O'Neill Estate B-1 / Exxon Mobil	Plugged, Dry Hole	5.27
South Fullerton Oil Co. #1	Idle, Oil & Gas	5.59

Corrected finding. Two plugged/abandoned exploratory wells lie within about one mile of the community centroid — one at ~0.25 miles, effectively within the footprint. A 2026 peer-reviewed California study reported a *suggestive, non-significant* association between proximity to **abandoned** wells within 10 km and childhood Ewing sarcoma (OR 1.27, 95% CI 0.96–1.66), stronger in Hispanic children. This places the community inside that exposure contrast and raises the priority of characterising these wells. It does **not** establish exposure or causation: these are mid-century plugged dry holes, and local groundwater is not used for supply.

5. Temporal relationships

PERIOD	LAND USE	CONFIDENCE
pre-1769	Acjachemen (Juaneño) native land	high
1840s-1882	Mexican land grant ranchos; cattle	high
1882-1940s	O'Neill/Flood partnership; cattle, wheat (largest OC wheat producer), row crops introduced	high
1940s-1998	Working cattle ranch; grazing dominant; barley dry-farming (pre-1983); lemon/citrus groves + frost wind machines; tree nurseries in canyon bottoms; stored agricultural chemicals	high (adjacent land) / moderate-high (exact footprint)
1997-1999	Ladera Planned Community entitled (OC ~1997, EIR 555); mass grading begins; first homes 1999	high
1999-2006	Ladera Ranch built out — 9 villages, ~8100 homes	high
2010	First delineated as Census CDP	high

The community was built out roughly 1999–2006 on land used for cattle grazing with pockets of dry-farmed barley, lemon/citrus orchard and tree nurseries. Children lived in Ladera Ranch during active mass grading of that former agricultural land — a mechanistically coherent route by which any legacy soil residue could have been mobilised as dust. Reported diagnoses span 2013–2026. The overlap is *plausible* and worth a residency-versus-construction timeline analysis; it is not evidence that any exposure occurred.

5.1 Does the timing actually work? Two mechanisms, opposite predictions

Grading occurred ~1999–2006; diagnoses span 2013–2026. Many affected children were not born when the earth was moved. That objection does real damage — but only to one of two mechanisms that have been getting conflated.

	M1 — CONSTRUCTION-ERA DUST	M2 — PERSISTENT SOIL RESIDUE
What it is	Grading mobilised buried residue as dust	Arsenic & organochlorines remain in surface soil
Time-bound?	Yes — an event, 1999–2006	No — a standing condition
Child present during grading?	Required (incl. in utero)	Not required
Predicts	Cases in birth cohorts ≤ ~2007	Cases across all birth cohorts
Effect of time passing	Weakens steadily	Does not weaken

Why M2 does not decay. Arsenic is a chemical element — it never degrades, and lead-arsenate orchard residues persist indefinitely. Organochlorines persist for decades, and the evidence here is local and empirical rather than theoretical: DTSC found DDT, toxaphene and arsenic in soil at former-farm school sites roughly three miles away, decades after cultivation ceased. A child born in 2012 playing in that soil has the same contact opportunity as one born in 2000.

The arithmetic on M1. A child could have been present during grading only if born on or before ~2007. The share of plausible pediatric ages (5–19) whose birth cohort overlaps grading falls from **93% for a 2013 diagnosis to 7% for a 2026 diagnosis**. By 2026 a case would have to be ~19 to have been present at all.

Net effect. "Grading mobilised dust and caused these cancers" is *substantially weakened* for recent diagnoses and unavailable for children born after ~2007. "The community sits on soil that may still contain arsenic and organochlorine residues, which children contact through ordinary play" is **untouched** by the timing objection. The leading hypothesis survives — in narrowed form.

The one datable Ewing case is a boundary case. Diagnosed August 2024 at age ~17 implies birth roughly Aug 2006 – Aug 2007 — either in utero during the final months of grading, or just after it ended. It cannot discriminate between M1 and M2. **Only one of the reported Ewing cases has both a published age and diagnosis year**, so no birth-cohort distribution can be built from public data. Since M1 and M2 predict opposite distributions, birth year is a *discriminating test* — and it is exactly what is missing. A further caveat cuts both ways: Ewing sarcoma's etiologic window is **unknown**, so no latency assumption can rule either mechanism in or out.

6. Observed versus expected — hypothetical scenarios

Person-years, ages 0–19, 2013–2026 $\approx$ 127,610; peak-age (10–19) $\approx$ 68,684. Expected = person-years $\times$ rate. SIR = observed $\div$ expected. Intervals are exact Poisson limits on the observed count.

SCENARIO	OBSERVED	RATE /M/YR	EXPECTED	SIR	95% CI
S1 central 6 cases, ages 0–19, all-races SEER rate	6	3.0	0.383	15.7	5.8 – 34.1
S2 ancestry-adjusted rate raised for a non-Hispanic-white-majority community	6	4.0	0.510	11.8	4.3 – 25.6
S3 peak-age ages 10–19 only, peak-age rate	6	4.6	0.315	19.1	7.0 – 41.5
S4 conservative count case-definition sensitivity	4	4.0	0.510	7.8	2.1 – 20.1
S5 higher count single-source outlier count	12	3.0	0.383	31.3	16.2 – 54.8
S6 leave-one-out one reported case removed	5	4.0	0.510	9.8	3.2 – 22.9

HYPOTHETICAL — not a finding. These scenarios use *unverified* public case reports and *estimated* population. They do not confirm a cluster and cannot substitute for an authorized California Cancer Registry analysis. Note that California suppresses any rate built on fewer than 15 cases or a population under 10,000, so a place-level Ewing sarcoma rate is **statistically unpublishable** — expected counts must be modelled from national rates, not queried locally. Compare S4 with S5 to see how severely a few cases move the result.

7. What is actually applied — the state's own pesticide data

We downloaded and processed California DPR's 2023 Pesticide Use Report archive (official machine-readable dataset, grade A1) — **79,473 Orange County application records** — to test, rather than assume, how much reported use can be placed on a map.

SITE TYPE	RECORDS	LOCATED	% LOCATED
Structural pest control	55,442	0	0.0%
Landscape maintenance	15,383	22	0.1%
Nursery — outdoor containers	2,990	2,982	99.7%
Golf course turf	1,375	0	0.0%
Rights of way	1,130	48	4.2%
Agriculture (fruiting pepper)	183	183	100.0%

This corrected an earlier assumption. Landscape maintenance *is* reported — 15,383 records, 110,664 lbs — but **99.9% of those records carry no township, range or section**. Overall, 94.6% of Orange County records have no location at all. Only agricultural and nursery categories are reliably geolocated. Separately, a federal cadastral lookup places Ladera Ranch in unsectioned former land-grant territory (T7S R7W / R8W, section 00). **Taken together, the state's pesticide reporting system is structurally incapable of placing an application inside Ladera Ranch.** That makes the HOA's own posted notices the only public location-specific evidence — and makes obtaining the HOA and vendor application logs the highest-value near-term request.

Top reported active ingredients, Orange County 2023

ACTIVE INGREDIENT	RECORDS	POUNDS
FIPRONIL	4,361	2,504
PIPERONYL BUTOXIDE	4,285	6,816
BIFENTHRIN	4,163	8,189
PYRIPROXYFEN	3,429	459
IMIDACLOPRID	3,390	3,043
DINOTEFURAN	3,042	5,314
PYRETHRINS	2,890	609
DIPHACINONE	2,854	7
INDOXACARB	2,585	83
DELTAMETHRIN	2,507	4,382
PRALLETHRIN	1,965	565
GLYPHOSATE, ISOPROPYLAMINE SALT	1,742	43,696

Glufosinate in context. The active ingredient named in Ladera Ranch common-area notices is independently confirmed in the state dataset: 442 Orange County records, 10,532 lbs, of which 336 records (10,177 lbs) were landscape maintenance. That corroborates the documented application pattern as **ordinary regional practice** — not as evidence of anything unusual, and not as evidence of causation. Glyphosate remains the larger landscape herbicide (~30,052 lbs).

8. Legacy soil arsenic — does a developed surface matter?

A fair objection: the community is fully built — grass, turf, concrete, asphalt. If legacy soil residue is the concern, is a child ever in contact with it? And if arsenic is in the soil, wouldn't lawn and backyard produce grown in that soil carry it anyway?

Arsenic has no half-life. It is element 33 and cannot degrade. ATSDR states it plainly: arsenic *"cannot be destroyed in the environment"* and *"tends to concentrate and remain in upper soil layers indefinitely."* The "6.5 to 16 year half-life" figures in circulation describe loss of the applied compound from the surface layer, not destruction of arsenic, and should not be used. This is the fundamental contrast with the herbicides in use here, which degrade in days to months.

	ARSENIC (LEGACY SOIL)	LANDSCAPE HERBICIDES
Persistence	Permanent — cannot degrade	Days to months (glufosinate ~7.4 d)
Exposure pattern	Continuous while resident	Repeated pulses, decaying between
Dominant route	Soil & house-dust ingestion	Dermal contact with treated turf; track-in
Effect of turf/hardscape	Barrier — reduces exposure	Source — is the contact surface
Established cancers	Bladder, lung, skin (adults, decades)	Glyphosate: NHL (IARC 2A); glufosinate: none
Ewing sarcoma link	None established	None established

The asymmetry that answers the question. For arsenic, turf is a barrier. For turf herbicides, turf is the source. Arsenic sits beneath the grass, so sod and hardscape put distance between a child and the affected layer. Herbicides are applied *onto* the grass children then play on. "It's all paved over" cannot be evaluated without first saying which hazard is being asked about.

What surface cover does not close: the indoor house-dust reservoir (EPA weights dust *above* soil — 60 vs 50 mg/day for ages 1–6 — and a randomised trial found yard coverings cut track-in ~50% while entryway dust lead did not significantly change over a year); gardening and digging; produce uptake; and disturbance events such as trenching or re-landscaping.

Backyard produce is real but crop-dependent. Soil-to-plant transfer is *more* efficient for arsenic than lead, ranking lettuce > carrot > bean > tomato. Fruiting vegetables can be grown safely even on substantially contaminated soil; leafy greens may exceed standards where soil arsenic is elevated.

But context reframes all of it. Southern California background arsenic — from 1,086 DTSC school-site samples across Orange and neighbouring counties — has a mean of **1.51 mg/kg** and an upper bound of **12 mg/kg**. The risk-based screening levels are **0.11–0.68 mg/kg**, i.e. roughly 18–110× *below* natural background. DTSC notes the risk-based number sits "100-times below typical ambient concentrations." **Exceeding a**

screening level in Orange County is normal, not evidence of contamination. Only a result above ~12 mg/kg would signify anything.

Revised standing: a LOW-PRIOR hypothesis. Six independent lines lower it. (1) Background swamps the screening levels. (2) Measured bioavailability at the closest analogue — Barber Orchard, a former lead-arsenate site — is **0.31**, so total soil arsenic overstates dose ~3x. (3) The crop history doesn't match: lead arsenate was an *apple and pear* insecticide, while this was citrus, barley and cattle land — any arsenical here more plausibly came from sodium-arsenite *herbicide* use. (4) Mass grading in 1999–2006 likely diluted the historic plough layer. (5) At Middleport NY, soil arsenic of 19.9 mg/kg produced **no significant correlation** with children's urinary arsenic ($r = 0.137$, $p = 0.39$); rice consumption was the significant predictor. (6) A systematic review found the childhood-cancer literature "does not seem to support an association between arsenic exposure and childhood cancers." **It is not excluded — it has never been measured here — but it should no longer be carried as the leading environmental explanation.**

9. Scientific literature

EVIDENCE DIRECTION	PAPERS
Does not resolve	5
Methodological/background	5
Supports pesticide hypothesis	5
Methodological	5
Weakens pesticide hypothesis	1
Weakens	1

Ewing sarcoma is a genetically and ancestry-driven, overwhelmingly sporadic cancer with **no established environmental cause**. Evidence tying pesticides specifically to Ewing sarcoma is weak, mixed and farming-*proxy* based: the strongest positive signals come from parental-farming studies with wide confidence intervals, while the largest and highest-quality occupational study (Great Britain, 5,369 sarcoma cases) found **no** pesticide or agriculture association. Two large California registry studies — on air pollution and on oil/gas proximity — were null overall. **No study links Ewing sarcoma to glyphosate, glufosinate, 2,4-D, or any specific pyrethroid.**

10. Alternative hypotheses that remain viable

Non-environmental

- Chance aggregation in a high-baseline-risk population.
- Germline susceptibility (GGAA-microsatellite / EWSR1 biology).
- Post-hoc boundary and ascertainment artifact.
- Exposure misclassification from residential mobility.

Environmental

- Legacy agricultural soil residue (highest priority, testable).
- Abandoned oil/gas well proximity.
- Construction-era soil disturbance and dust.
- Recycled irrigation water (under-characterised).
- Traffic-related air pollution (SR-241 corridor).

11. Claims that remain unproven

- Causation by any exposure.
- That a true epidemiological cluster exists.
- The specific role of glufosinate or any named chemical.
- The "17 pesticides" figure and the higher "12 Ewing" case count.
- That legacy soil residue is present on the residential footprint.
- That recycled irrigation water carries any hazard.

12. What would most improve the analysis

1. **Registry-confirmed case data** from the California Cancer Registry — counts, diagnosis dates, ages, ancestry, and critically **birth year**. Birth year alone would discriminate between the two leading environmental mechanisms (see §5.1). Highest ability to change conclusions.
2. **Individual residence histories** across the etiologic window (consented, IRB-governed).
3. **Soil sampling** of the footprint and common areas for arsenic, DDT/DDE and toxaphene — the decisive test of the leading environmental hypothesis.
4. **HOA and vendor application logs** — now the *only* possible source of location-specific application data, and at risk of routine disposal.
5. **The entitlement/EIR soil-testing record** — did anyone ever test this land before building on it?

13. Evidence quality

GRADE	MEANING	COUNT
A1	Official dataset / peer-reviewed / registry	20
A2	Official government webpage, filing, meeting doc	18
B1	University / research-institution report	9
B2	Reputable news quoting named sources	8
C	Advocacy, law-firm, petition, unverified counts	11
D	Speculation / unsourced reposts	2

Lower-grade sources are retained as leads and never silently promoted. Where sources conflict — as they do on case counts — both values are recorded with their grades, and the discrepancy is treated as a finding about data quality rather than smoothed away.

Correction log. An inherited study centroid was found to be ~1.93 miles too far north; it was caught when this report's map figure showed the study zone sitting north of the community. All site and well distances were recomputed. Former-agricultural sites are closer than previously recorded, and two abandoned wells moved from ~2–2.7 miles to ~0.25 and ~0.77 miles. Full record: [research/CORRECTIONS.md](#).

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